WORKED SOLUTION

STEP II 2021, Question 8

Pure • Integral Recurrence Relations

STEP II 2021, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.9, *Fractions That Close In on e*: https://beyondstep.keithmcnulty.org/ch02/e-approximations.html

The key idea

The function $t^n(1-t)^n$ is a power of the single quadratic $u = t(1-t)$, so differentiating it is just the chain rule twice. That function and its first derivative both vanish at $t = 0$ and $t = 1$, so integrating by parts twice against e^t throws away every boundary term and turns the second derivative from part (i) straight into a recurrence. The integrals are squeezed between 0 and something tiny, and this forces e to be very well approximated by the fractions $-a_n/b_n$.

Solution

Part (i)

Let $u = t(1-t) = t - t^2$, so that $\dfrac{\mathrm{d}u}{\mathrm{d}t} = 1 - 2t$, $\dfrac{\mathrm{d}^2u}{\mathrm{d}t^2} = -2$, and

$$(1-2t)^2 = 1 - 4t + 4t^2 = 1 - 4u.$$

With $f(t) = t^n(1-t)^n = u^n$, the chain rule gives $f'(t) = nu^{n-1}(1-2t)$. Differentiating again by the product rule (valid for $n \geq 2$, so that u^{n-2} is a polynomial),

$$\begin{aligned} f''(t) &= n(n-1)u^{n-2}(1-2t)^2 - 2nu^{n-1} \\ &= nu^{n-2}\big[(n-1)(1-4u) - 2u\big] \\ &= nu^{n-2}\big[(n-1) - (4n-2)u\big] \\ &= nt^{n-2}(1-t)^{n-2}\big[(n-1) - 2(2n-1)t(1-t)\big], \end{aligned}$$

which is the required result.

Part (ii)

Keep $f(t) = t^n(1-t)^n$ with $n \geq 2$, so $n!\,T_n = \int_0^1 f(t)\,\mathrm{e}^t\,\mathrm{d}t$. Integrating by parts twice, each time differentiating the polynomial and integrating e^t,

$$\int_0^1 f\,\mathrm{e}^t\,\mathrm{d}t = \Big[f\,\mathrm{e}^t\Big]_0^1 - \Big[f'\,\mathrm{e}^t\Big]_0^1 + \int_0^1 f''\,\mathrm{e}^t\,\mathrm{d}t.$$

? **Ask yourself** Why do both boundary terms disappear?

Because $f(t) = u^n$ and $f'(t) = nu^{n-1}(1-2t)$ both contain a factor $u = t(1-t)$ (here we use $n \geq 2$, so $n-1 \geq 1$), and $u = 0$ at $t = 0$ and at $t = 1$. So

$$n!\,T_n = \int_0^1 f''(t)\,\mathrm{e}^t\,\mathrm{d}t.$$

Now substitute part (i), writing it as $f'' = n(n-1)u^{n-2} - 2n(2n-1)u^{n-1}$:

$$\begin{aligned} n!\,T_n &= n(n-1)\int_0^1 u^{n-2}\mathrm{e}^t\,\mathrm{d}t - 2n(2n-1)\int_0^1 u^{n-1}\mathrm{e}^t\,\mathrm{d}t \\ &= n(n-1)\,(n-2)!\,T_{n-2} - 2n(2n-1)\,(n-1)!\,T_{n-1} \\ &= n!\,T_{n-2} - 2(2n-1)\,n!\,T_{n-1}, \end{aligned}$$

using $\int_0^1 u^k\mathrm{e}^t\,\mathrm{d}t = k!\,T_k$. Dividing by $n!$ gives

$$T_n = T_{n-2} - 2(2n-1)T_{n-1} \qquad (n \geq 2),$$

as required.

Part (iii)

First, $T_0 = \int_0^1 \mathrm{e}^t\,\mathrm{d}t = \mathrm{e} - 1$.

For $T_1 = \int_0^1 (t - t^2)\mathrm{e}^t\,\mathrm{d}t$ we can check by differentiating that $\int t\,\mathrm{e}^t\,\mathrm{d}t = (t-1)\mathrm{e}^t$ and $\int t^2\mathrm{e}^t\,\mathrm{d}t = (t^2 - 2t + 2)\mathrm{e}^t$, so

$$\int_0^1 (t - t^2)\mathrm{e}^t\,\mathrm{d}t = \Big[(-t^2 + 3t - 3)\mathrm{e}^t\Big]_0^1 = -\mathrm{e} + 3,$$

that is, $T_1 = 3 - \mathrm{e}$.

So $T_0 = a_0 + b_0\mathrm{e}$ with $(a_0, b_0) = (-1, 1)$, and $T_1 = a_1 + b_1\mathrm{e}$ with $(a_1, b_1) = (3, -1)$, all integers. Suppose that T_{n-2} and T_{n-1} have this form with integer coefficients, for some $n \geq 2$. By part (ii),

$$T_n = \big(a_{n-2} - 2(2n-1)a_{n-1}\big) + \big(b_{n-2} - 2(2n-1)b_{n-1}\big)\mathrm{e},$$

and both brackets are integers because $2(2n-1)$ is an integer. By (strong) induction, every T_n with $n \geq 0$ can be written as $a_n + b_n\mathrm{e}$ with $a_n, b_n \in \mathbb{Z}$.

Part (iv)

The bounds. For $0 < t < 1$ we have $0 < t(1-t) \le \frac{1}{4} < 1$, so $0 < t^n(1-t)^n \le 1$, and also $0 < \mathrm{e}^t < \mathrm{e}$. Hence on the open interval $(0,1)$

$$0 < \frac{t^n(1-t)^n}{n!}\,\mathrm{e}^t < \frac{\mathrm{e}}{n!}.$$

The integrand is continuous, so integrating these strict inequalities over $[0,1]$ (an interval of length 1) keeps them strict:

$$0 < T_n < \frac{\mathrm{e}}{n!} \qquad (n \ge 0).$$

(For $n = 0$ the left factor is just 1, and the strictness comes from $\mathrm{e}^t < \mathrm{e}$.)

The limit. Since $b_n \neq 0$, divide $T_n = a_n + b_n\mathrm{e}$ by b_n:

$$-\frac{a_n}{b_n} - \mathrm{e} = -\frac{T_n}{b_n}, \qquad \text{so} \qquad \left|-\frac{a_n}{b_n} - \mathrm{e}\right| = \frac{T_n}{|b_n|} \le T_n < \frac{\mathrm{e}}{n!}.$$

Here we used that b_n is a non-zero *integer*, so $|b_n| \ge 1$. As $n \to \infty$, $\mathrm{e}/n! \to 0$, so $-a_n/b_n \to \mathrm{e}$.

In fact the bound is very generous. From the recurrence, $(a_2, b_2) = (-19, 7)$ and $(a_3, b_3) = (193, -71)$, giving the approximations $\frac{19}{7} \approx 2.714$ and $\frac{193}{71} \approx 2.71831$, against $\mathrm{e} \approx 2.71828$.

Checks

- Put $n = 2$ in part (i): $t^2(1-t)^2 = t^2 - 2t^3 + t^4$ has second derivative $2 - 12t + 12t^2$, and the formula gives $2\big[1 - 6t(1-t)\big] = 2 - 12t + 12t^2$.
- $T_1 = 3 - \mathrm{e} \approx 0.2817$ is positive and less than $\mathrm{e}/1!$, as part (iv) says it must be. Then $T_2 = T_0 - 6T_1 = 7\mathrm{e} - 19 \approx 0.0280$, which a computer evaluation of the integral confirms, along with the recurrence and the bounds for n up to 11.
- The values $-a_n/b_n$ for $n = 4, 5, 6$ are $\frac{2721}{1001}$, $\frac{49171}{18089}$, $\frac{1084483}{398959}$, which differ from e by about 10^{-7}, 3×10^{-10} and 5×10^{-13}: much faster than $\mathrm{e}/n!$ alone would suggest.

Common slips

- Forgetting to say why the boundary terms vanish, or not noticing that this needs $n \ge 2$: for $n = 1$ the derivative $1 - 2t$ is non-zero at both ends.
- Using only $b_n \neq 0$ in part (iv) without $|b_n| \ge 1$. A non-zero real number could be tiny; it is the fact that b_n is an integer that stops T_n/b_n from blowing up.

Beyond STEP

This is the heart of Hermite's 1873 approach to proving that e is irrational and even transcendental: build integer combinations $a_n + b_n\mathrm{e}$ that are non-zero but tend to zero. If $\mathrm{e} = p/q$, then $q(a_n + b_n\mathrm{e}) = qa_n + b_n p$ would be a positive integer less than $q\mathrm{e}/n!$, which is impossible for large n; the same polynomials $t^n(1-t)^n$ appear in Beukers' famous proof that $\zeta(3)$ is irrational. For more on how well numbers can be approximated by fractions, read Davenport, *The Higher Arithmetic*.

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